```
print("All libraries imported successfu
```

```
All libraries imported successfully!
```

...

	Region	Date	Frequency	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	Area
0	Andhra Pradesh	31-05-2019	Monthly	3.65	11999139.0	43.24	Rural
1	Andhra Pradesh	30-06-2019	Monthly	3.05	11755881.0	42.05	Rural
2	Andhra Pradesh	31-07-2019	Monthly	3.75	12086707.0	43.50	Rural
3	Andhra Pradesh	31-08-2019	Monthly	3.32	12285693.0	43.97	Rural
4	Andhra Pradesh	30-09-2019	Monthly	5.17	12256762.0	44.68	Rural

```
 #   Column                                   Non-Null Count  Dtype
---  ------                                   --------------  -----
 0   Region                                   740 non-null    object
 1    Date                                    740 non-null    object
 2    Frequency                               740 non-null    object
 3    Estimated Unemployment Rate (%)         740 non-null    float64
 4    Estimated Employed                      740 non-null    float64
 5    Estimated Labour Participation Rate (%) 740 non-null    float64
 6   Area                                     740 non-null    object
dtypes: float64(3), object(4)
memory usage: 42.1+ KB
```

	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)
count	740.000000	7.400000e+02	740.000000
mean	11.787946	7.204460e+06	42.630122
std	10.721298	8.087988e+06	8.111094
min	0.000000	4.942000e+04	13.330000
25%	4.657500	1.190404e+06	38.062500
50%	8.350000	4.744178e+06	41.160000
75%	15.887500	1.127549e+07	45.505000
max	76.740000	4.577751e+07	72.570000

```
Missing Values:
Region                                   28
 Date                                    28
 Frequency                               28
 Estimated Unemployment Rate (%)         28
 Estimated Employed                      28
 Estimated Labour Participation Rate (%) 28
Area                                     28
dtype: int64
```

	Region	Date	Frequency	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	Area
0	Andhra Pradesh	31-05-2019	Monthly	3.65	11999139.0	43.24	Rural
1	Andhra Pradesh	30-06-2019	Monthly	3.05	11755881.0	42.05	Rural
2	Andhra Pradesh	31-07-2019	Monthly	3.75	12086707.0	43.50	Rural

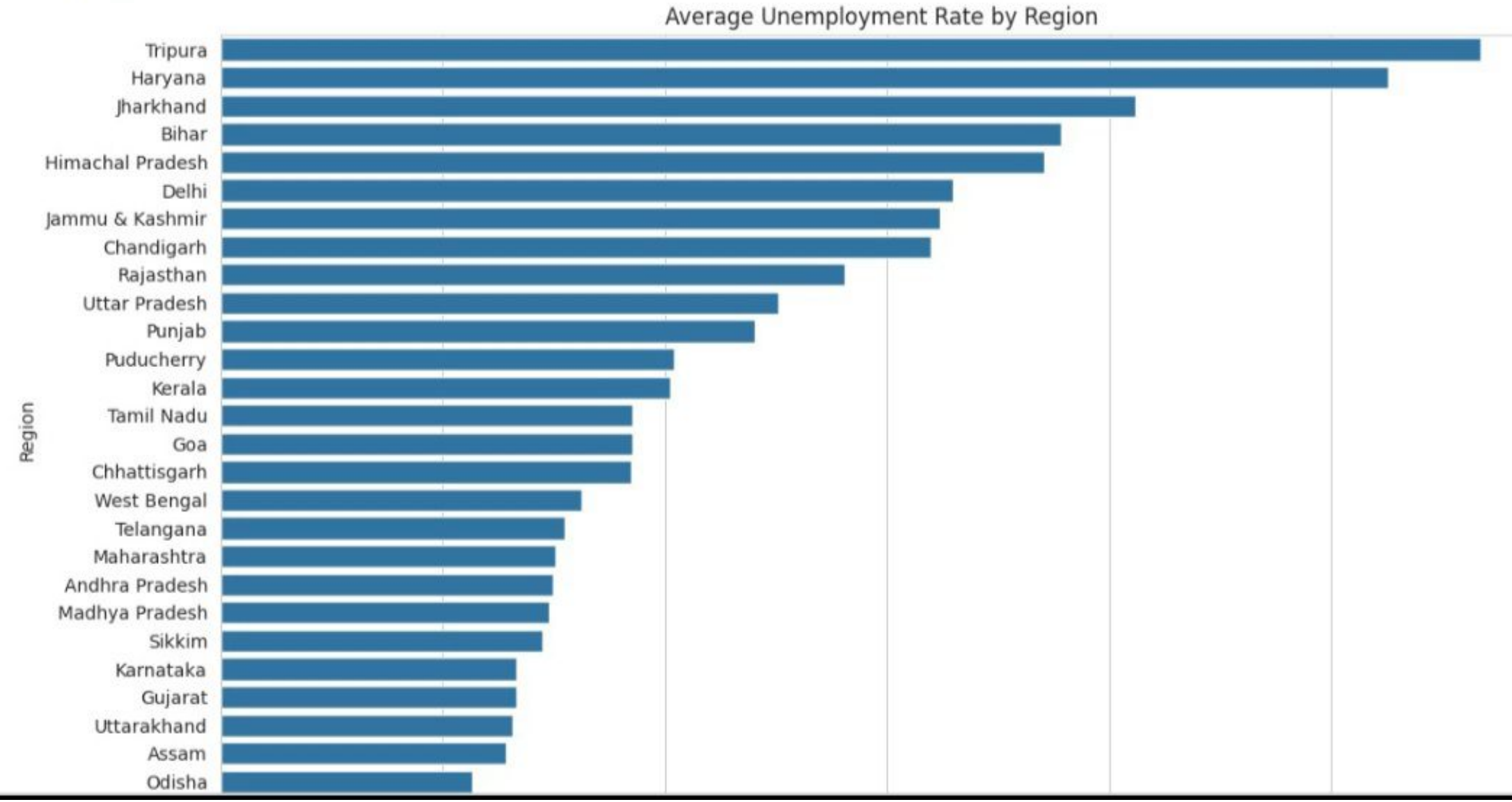
```
display(df.head())
```

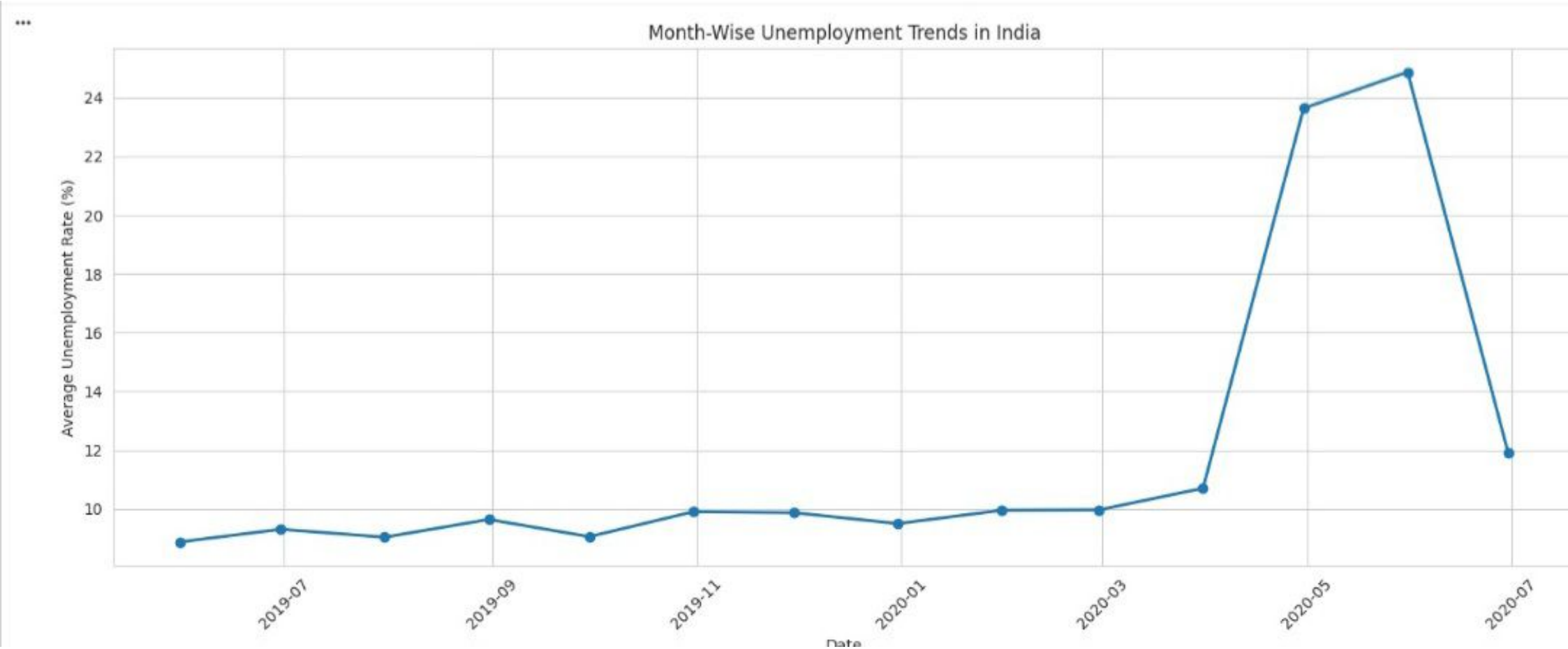
```
Missing values after conversion:
Region                        28
Date                          28
Frequency                     28
Unemployment_Rate             28
Employed                      28
Labour_Participation_Rate     28
Area                          28
dtype: int64

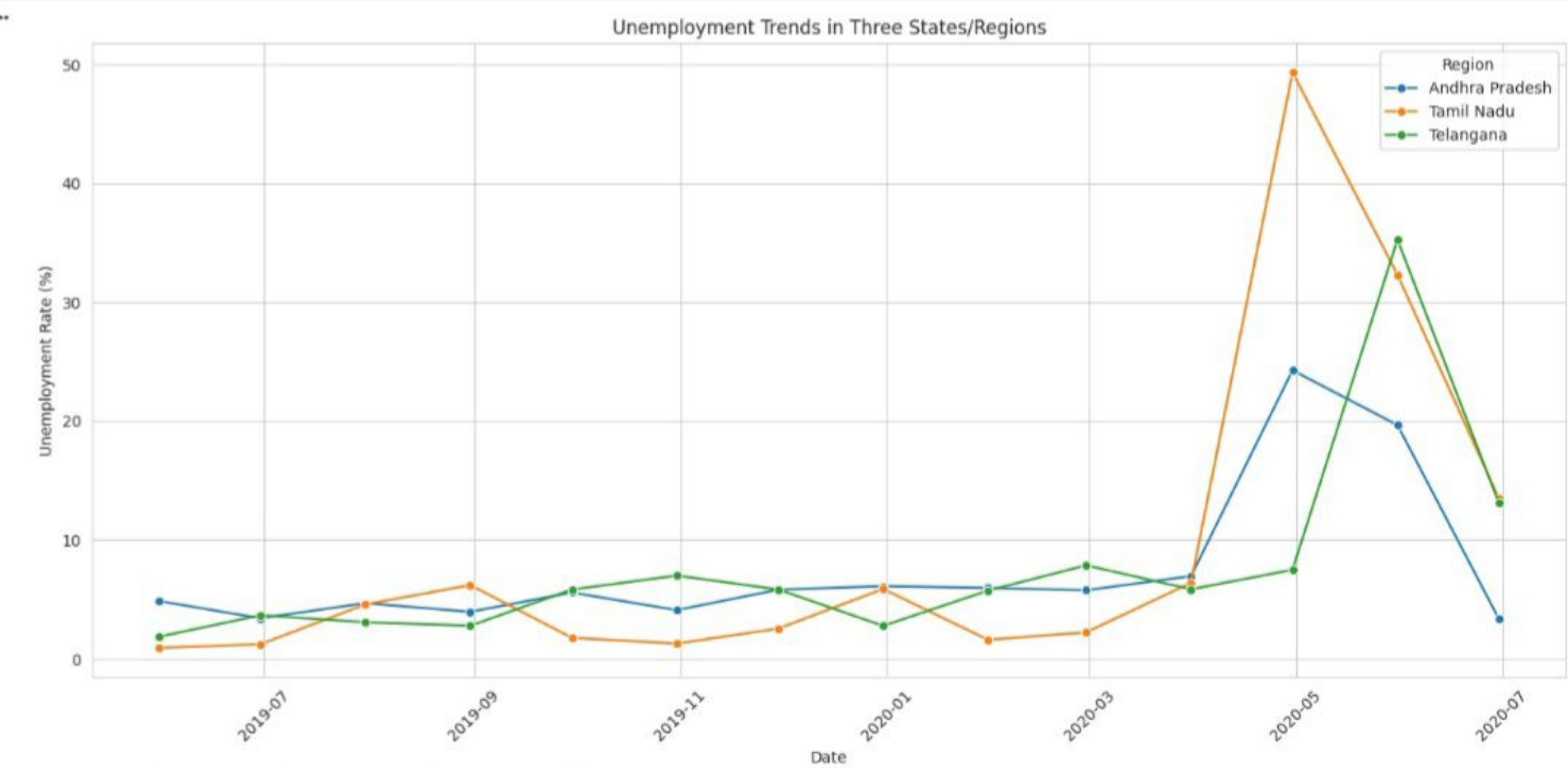
Dataset cleaned successfully!
Cleaned dataset shape: (740, 7)
```

	Region	Date	Frequency	Unemployment_Rate	Employed	Labour_Participation_Rate	Area
0	Andhra Pradesh	2019-05-31	Monthly	3.65	11999139.0	43.24	Rural
1	Telangana	2019-05-31	Monthly	2.23	11053353.0	61.74	Rural
2	Tamil Nadu	2019-05-31	Monthly	0.97	15844698.0	49.44	Rural
3	Rajasthan	2019-05-31	Monthly	4.03	15226005.0	38.52	Rural
4	Punjab	2019-05-31	Monthly	9.17	6088547.0	44.79	Rural

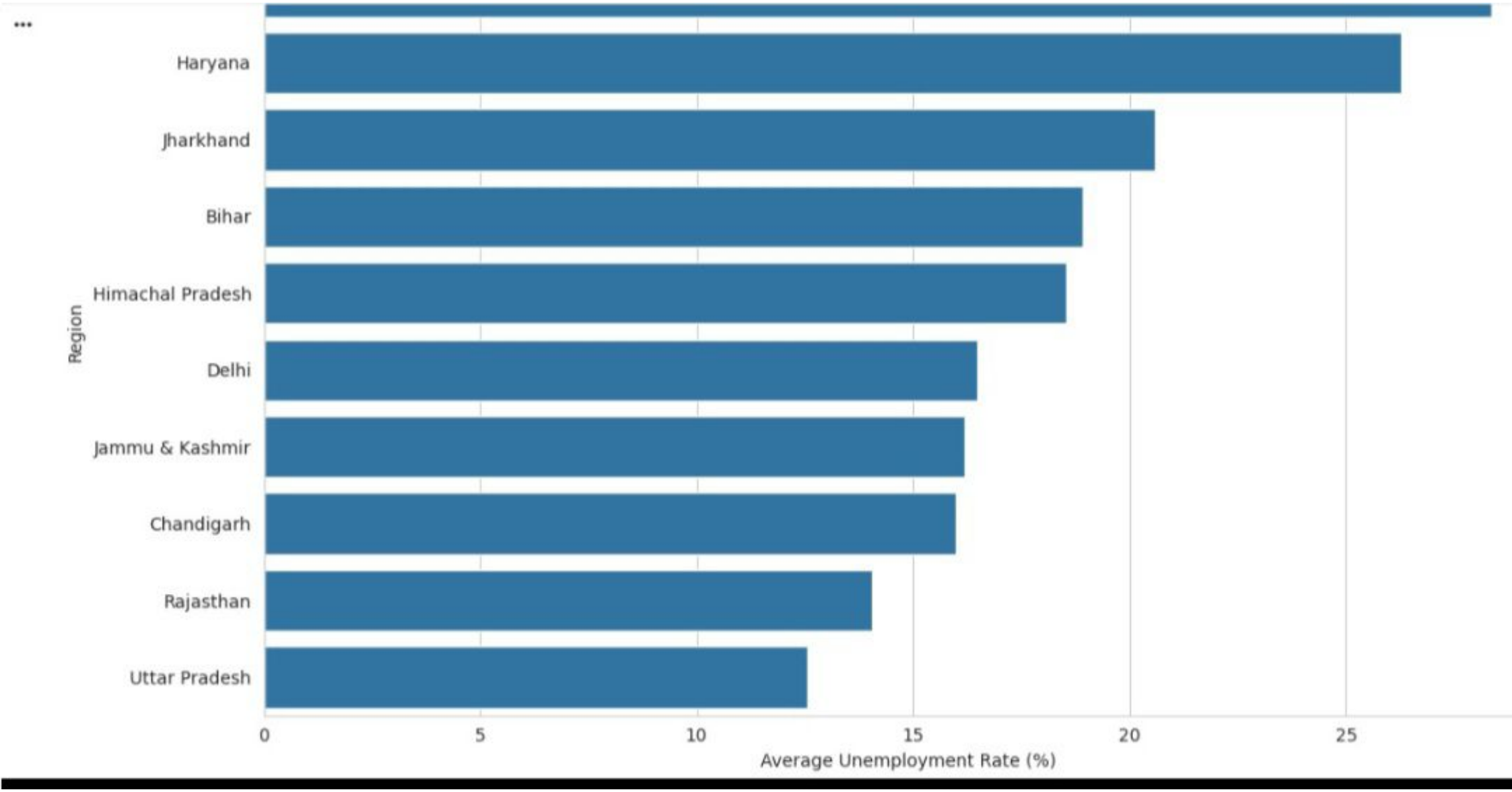
Gujarat	6.663929
Uttarakhand	6.582963
Assam	6.428077
Odisha	5.657857
Meghalaya	4.798889



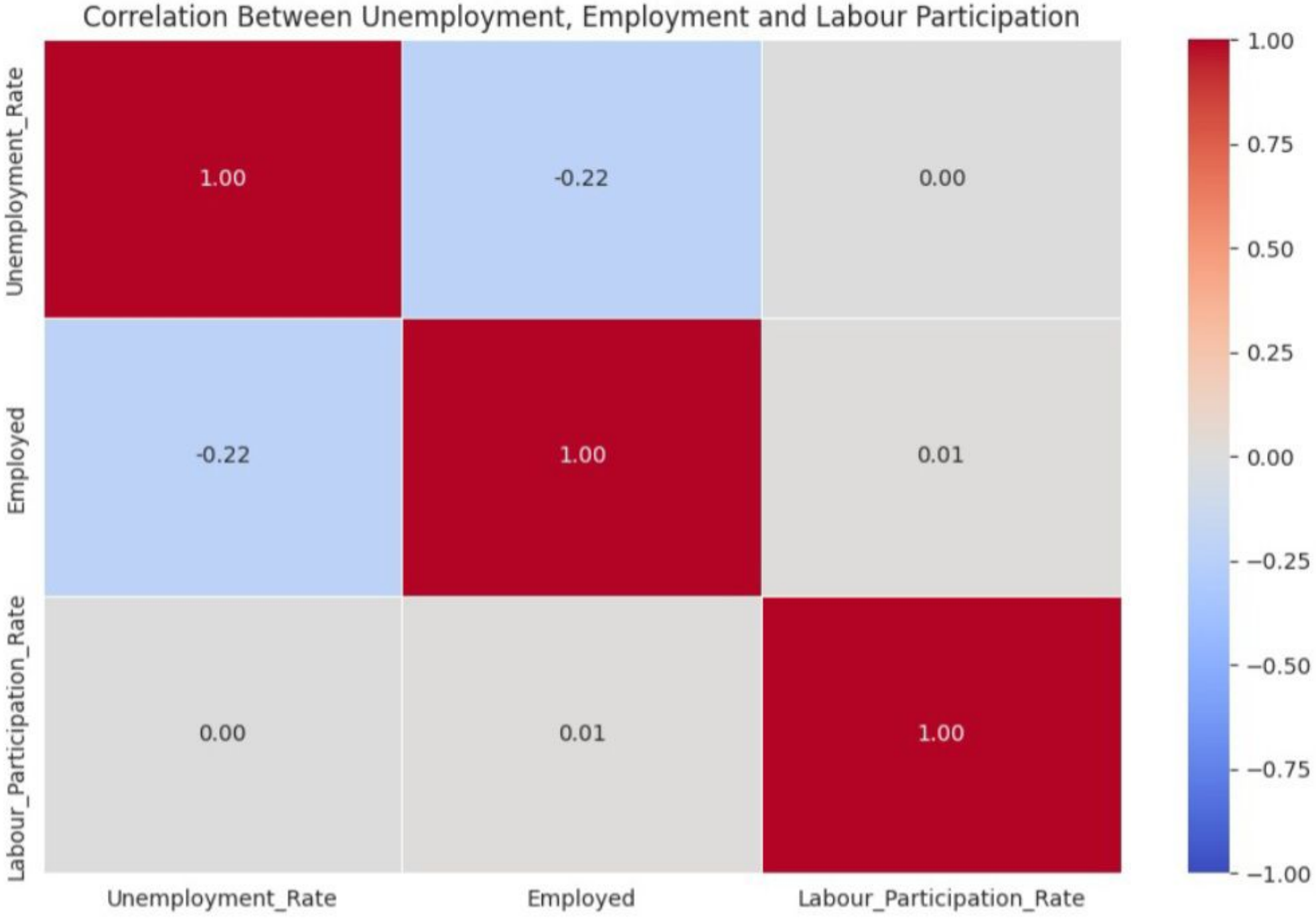

Month-Wise Unemployment Trends in India
Average Unemployment Rate (%)
24
22
20
18
16
14
12
10
2019-07
2019-09
2019-11
2020-01
2020-03
2020-05
2020-07
Date



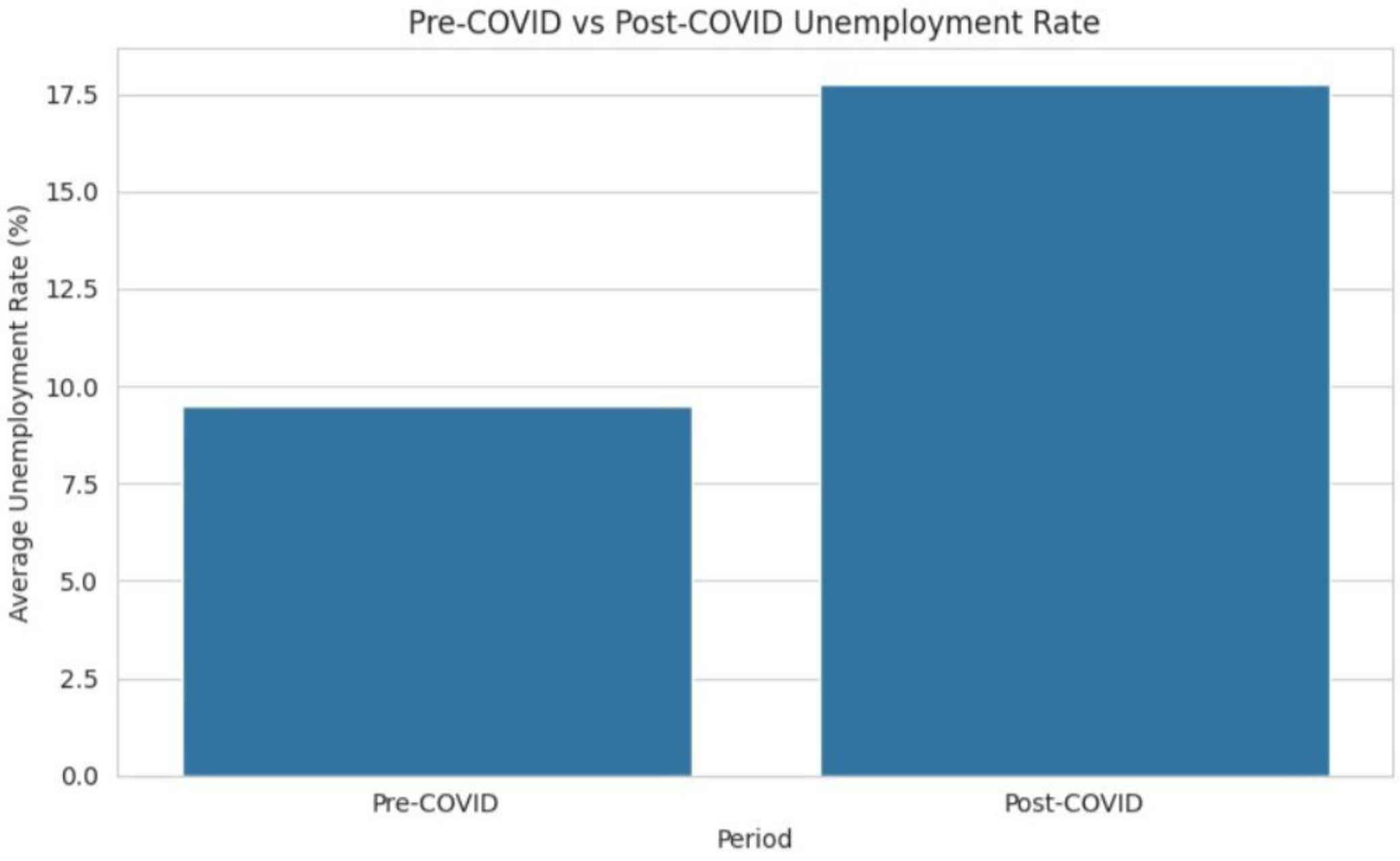
States/regions included: ['Andhra Pradesh', 'Telangana', 'Tamil Nadu']


Haryana
Jharkhand
Bihar
Himachal Pradesh
Delhi
Jammu & Kashmir
Chandigarh
Rajasthan
Uttar Pradesh
Region
0
5
10
15
20
25
Average Unemployment Rate (%)

Employed	-0.222876	1.000000	0.011300
Labour_Participation_Rate	0.002558	0.011300	1.000000

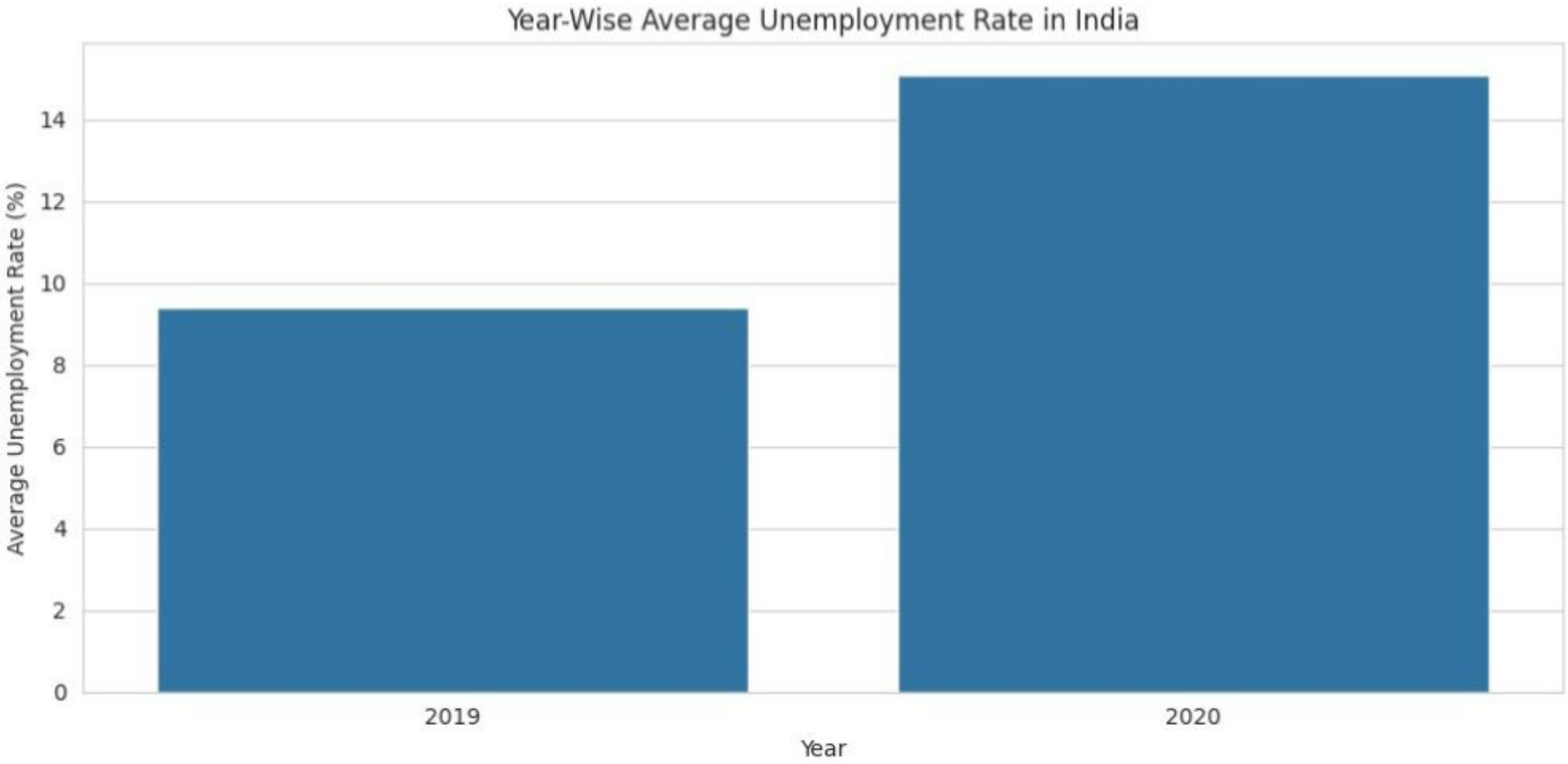


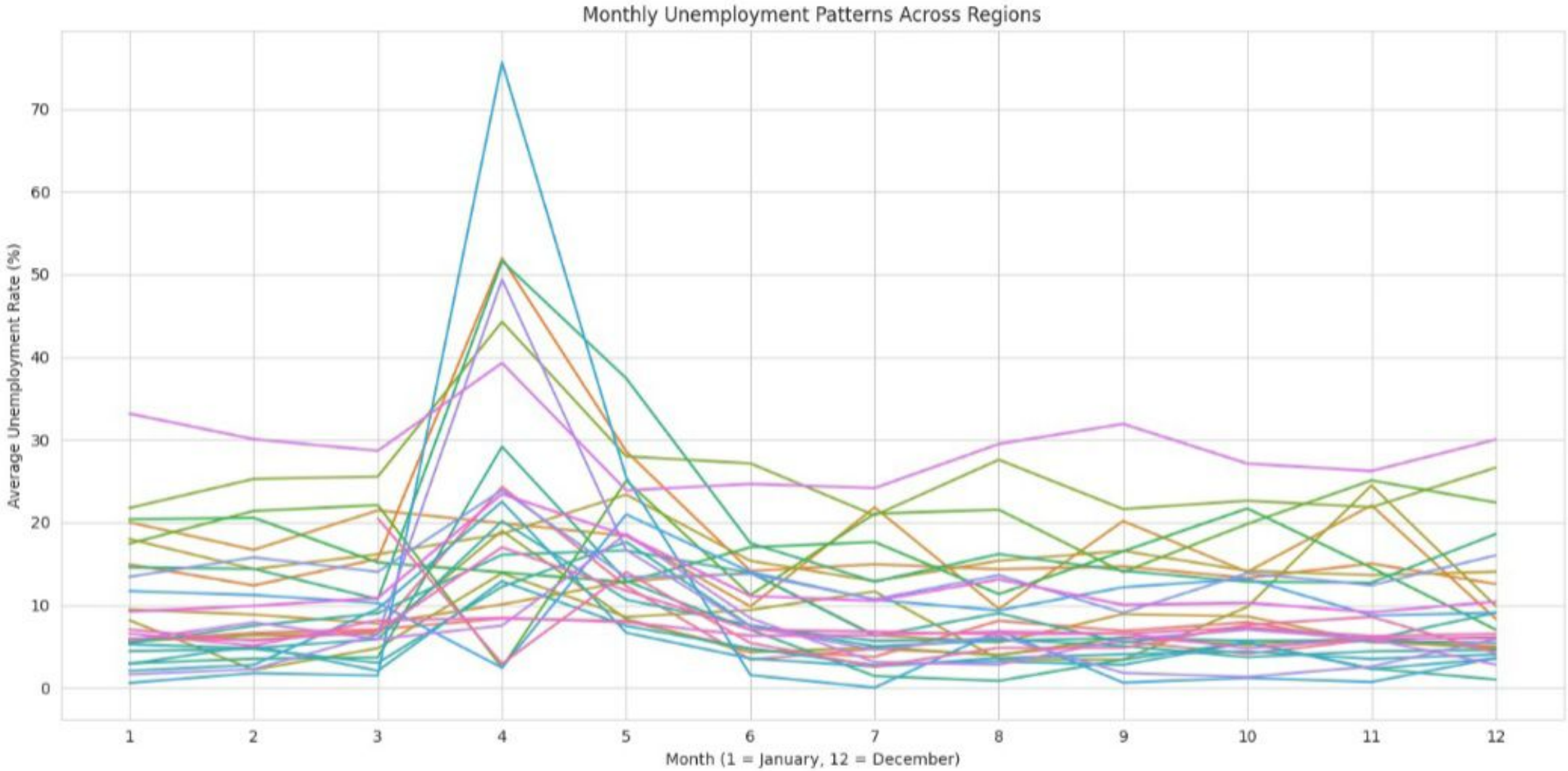
Pre-COVID	9.509534
Post-COVID	17.774363



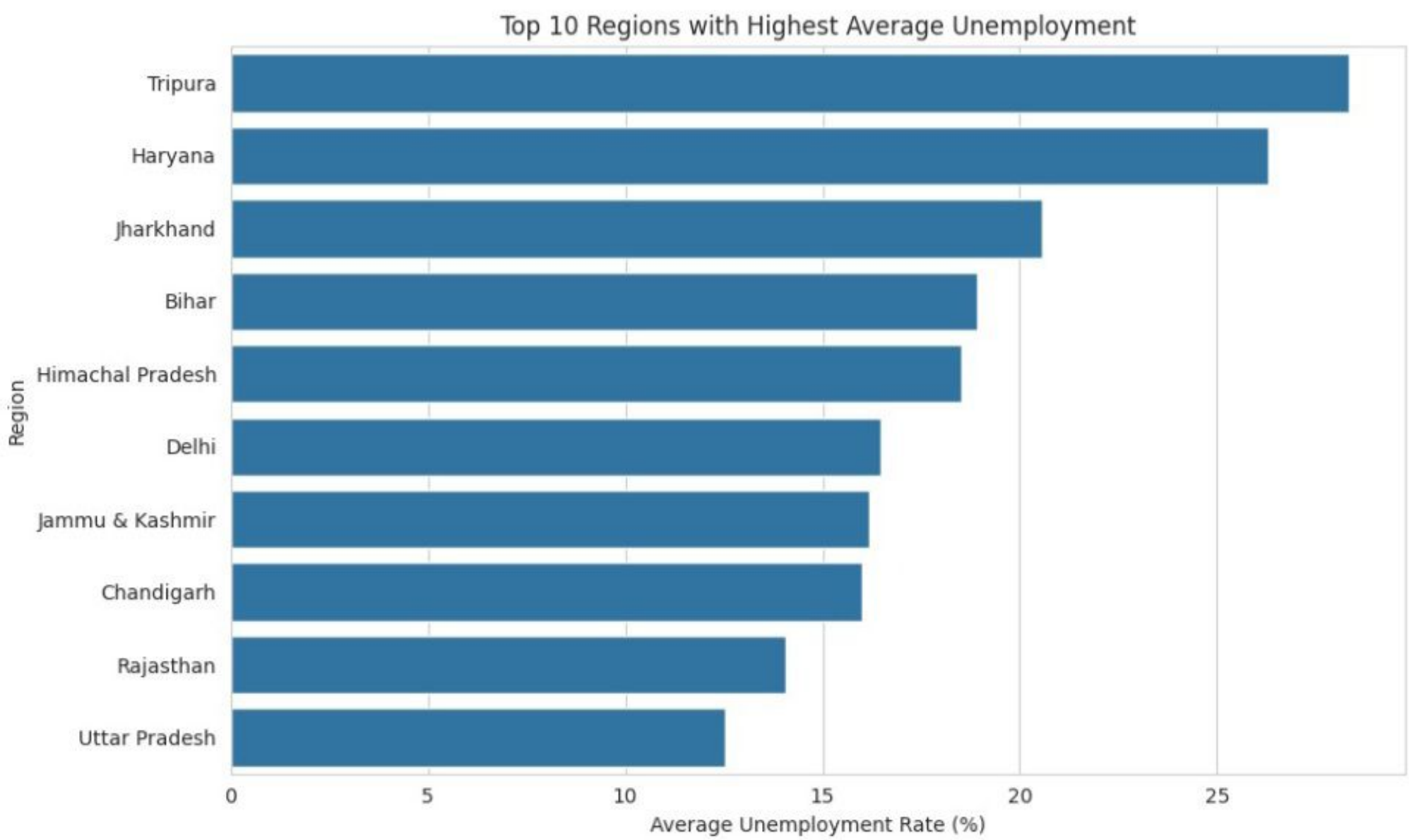

```
Difference in average unemployment rate: 8.26 percentage points
The average unemployment rate was higher in the post-COVID period.
```

Year	
2019	9.399047
2020	15.101581




Monthly Unemployment Patterns Across Regions
Average Unemployment Rate (%)
0
10
20
30
40
50
60
70
1
2
3
4
5
6
7
8
9
10
11
12
Month (1 = January, 12 = December)

```
Cleaned dataset saved successfully!
```


Top 10 Regions with Highest Average Unemployment
Region
Tripura
Haryana
Jharkhand
Bihar
Himachal Pradesh
Delhi
Jammu & Kashmir
Chandigarh
Rajasthan
Uttar Pradesh
0
5
10
15
20
25
Average Unemployment Rate (%)